**UID EX 6**

**RAD** (Rapid Application Development) is a software development methodology that emphasizes **quick development and iteration** of prototypes over strict planning and long development cycles. It's ideal for projects where requirements evolve through user feedback and where fast delivery is important.

**Key Characteristics of RAD:**

- Focus on **prototypes** instead of extensive documentation
- **User involvement** throughout the development process
- **Fast development cycles** with frequent iterations
- **Reusable components** to reduce effort
- Best suited for **UI/UX-intensive projects** like web and mobile apps

**<u>Phase 1</u>: Requirements Planning**

This is where the **project goals, system requirements, and user needs** are identified. Stakeholders, users, and designers collaborate to decide what needs to be built.

**1) Identify Key Features**

**Navigation:**

- Home
- Product Categories
- Product Listings
- Product Details
- Cart
- Order Confirmation

**User Actions:** Browsing, Searching, Adding to Cart, Checkout, Tracking Orders

**2) Create a Requirements Document:**

1. View list of product categories
2. View products under each category
3. View detailed information about a product
4. View and edit cart
5. Checkout and enter user/shipping/payment details
6. View order confirmation message

**<u>Phase 2</u>: User Design**

Designers and users collaborate to **create wireframes and prototypes**. The idea is to show how the application will look and behave before any actual coding starts.

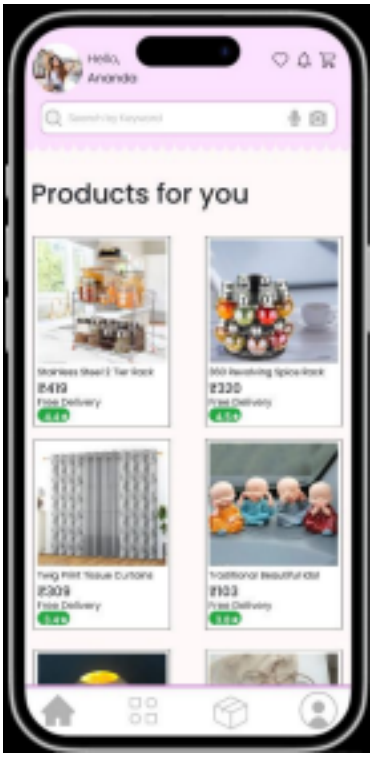
**1) Tools Used: Axure**: Used to design the wireframes and UI prototype.

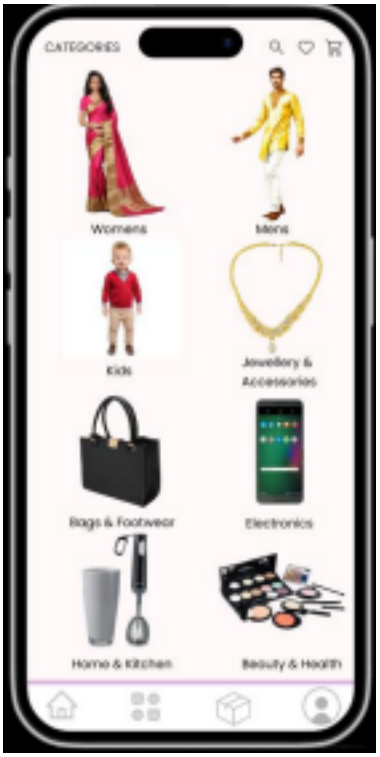
**2) Wireframes Designed:**

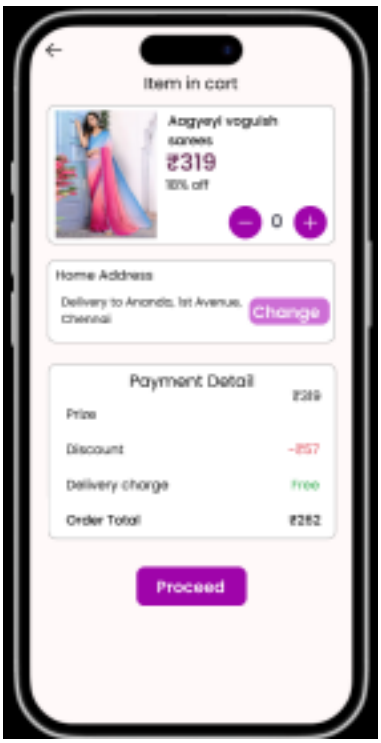
- **Home Page:** Highlights featured products and links to categories.
- **Product Categories:** Lists all product categories to explore.
- **Product Listings:** Shows products under a selected category.
- **Product Details:** Displays product info with "Add to Cart" option.

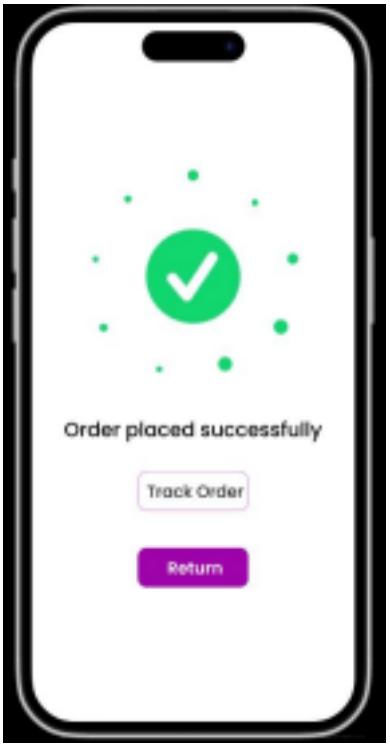
**Cart:** Shows added items with price and quantity details.
• **Checkout:** Collects shipping and payment information. •
**Order Confirmation:** Confirms successful order placement.

Designed **interactions and navigation** using Axure's Prototype mode
Reused components like headers/footers using **Components** Annotated elements for clarity (e.g., what a button does, purpose of sections)





**<u>Phase 3</u>: Construction**

This is where the **actual development happens**, but in RAD, it is tightly coupled with ongoing testing and feedback. You refine the prototype based on user responses.

**1) Interactive Prototypes:**

- Created using Axure's **prototype linking** features.
- Functional paths designed:
  - From **Home → Product Categories → Product Listings → Product Details → Add to Cart → Checkout → Order Confirmation**

**2) Dynamic Elements:**

- Simulated transitions and modals for adding to cart and checking out. • Screens simulate real-time behavior such as navigating between product listings and cart.

**3) Testing and Iteration:**

- Prototype previewed with **"Present"** mode in Figma.
- Collected feedback from stakeholders (assumed for documentation). • Iterated based on visual hierarchy, accessibility, and flow suggestions.

**<u>Phase 4</u>: Cutover**

In this final phase, the system goes live. This includes **finalizing the design**, **training users**, and **preparing documentation** for developers.

**1) Finalization and Export:**

- Final design prototype shared through Figma link.
- Ready for hand-off to development team.
- Could be exported to **HTML/CSS** using plugins like Figma-to-HTML or further tools if needed.

**2) User Training & Support:**

- Walkthrough of the prototype for stakeholders or development team. • Notes/documentation provided for developers to understand functionality and user flow.
- Easy access to Figma comments for further collaboration and questions.